# Robed Beauvile-Owen, Ph.D.

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## EXPERIENCE

**Founder & Data Scientist,** *AniPoint Site Research* **(anipointresearch.com)** Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy, identifying $1.5M+ in expansion revenue potential for clients from site-level car-count projections.
- Built a revenue driver tree (traffic × capture rate × average ticket) and margin models that turned client profit goals into break-even targets.
- Estimated the incremental revenue of adding a site to a county (holding other factors constant) with Double ML and causal forests on a 12-year county panel.
- Cross-checked with a two-way fixed-effects DiD regression on past county changes, which agreed with the Double ML estimate within 20% ($329K vs. $399K per added site).
- Automated Python pipelines for Monte Carlo risk and low/base/high scenarios (150+ unit tests, data-quality checks).

**Senior Data Scientist,** *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Developed uplift and segmentation models on millions of HIPAA-protected records, used in 87% of campaigns, reducing per-campaign spend by $10K while maintaining baseline enrollment volumes.
- Built an in-house experimentation and reporting platform that automated A/B test design and readouts (power analysis, CUPED variance reduction, significance tests), metric monitoring, and Power BI dashboards for data science and strategy.
- Identified insurance claims and support-ticket complaints as the strongest leading signals of member churn and added them to the churn model, raising its PR-AUC by 5% and AUC-ROC by 2% (relative).
- Revamped the annual retention survey analysis in R and automated it end to end, cutting turnaround from 2.5 weeks to 1 day.
- Re-architected the churn pipeline on Spark and Databricks (partitioning, caching, removing repeated joins), cutting runtime by 48% and memory by 23%.
- Built a certified gross lead rate (GLR) definition for a new client whose data lacked key lead markers, working with data engineering to date leads by last response, credit them to campaigns, and flag broker-influenced leads.

**Founding Machine Learning Engineer,** *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- Built a regression shipping price model, validated it offline (MAE, human-in-the-loop review) and in A/B tests, and shipped it as a pricing API (C#, Azure, 10,000+ daily requests) that replaced a heuristic rule, cutting erroneous and outlier prices by 13% (relative) while maintaining an 18% contribution margin.
- Built automated drift monitoring for live inference pipelines and mentored the software engineering team on production ML practices.
- Cut search complaints by 12% (relative) and raised search click-through rate by 16% by tuning Elasticsearch relevance (custom analyzers, field boosting), validated with NDCG@10, human review, and A/B tests.

**Data Scientist,** *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Improved the XGBoost GLR model's offline AUC-ROC by 3% relative to the prior model and used SHAP to identify the main drivers (income, age, household composition).
- Confirmed the gain in an A/B test, where model-driven targeting raised GLR by 9% (relative).
- Estimated the lift in GLR caused by new creatives with uplift models (treatment vs. control), and validated GLR out of sample as a 60-day leading indicator of enrollment rate ($R^2 = 0.28$).
- Raised the precision of high-opportunity zone selection by 20% (relative) over legacy zoning (scored against past successful campaigns in similar locations), using spatial overlays and demographic indexing in GeoPandas.
- Led monthly monitoring of up to 10 production client models (accuracy, feature drift, forecast vs. actual) and built a company-wide self-service dashboard on curated Snowflake and Databricks tables.

## TECHNICAL PROJECTS

**Customer LTV & Churn Survival Modeling** github.com/beauvilerobed/.../projects/ltv-churn-survival

- Simulated 1M customer lifetimes with churn falling over tenure (10%, 5%, then 2% monthly) to show why the standard LTV = ARPU / churn formula misleads unless churn is constant.
- Estimated expected lifetime and LTV from the empirical survival curve ($E[T] = \int S(t)dt$), which recovered the simulated mean.

**Medicare Claims Intelligence & Fraud Detection** github.com/beauvilerobed/.../projects/medicare-claims-intelligence

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K CMS claims, with tests that caught 11,253 claims missing a service date.
- Beat a seasonal-naive forecast in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), reached 81% precision on planted fraud at review capacity, and built an out-of-time XGBoost risk model (SHAP, MLflow) finding 46% more high-cost members than a prior-year-cost baseline.

## EDUCATION

**Ph.D. in Mathematics,** *University of Pittsburgh* 2017 - 2023

**Dissertation:** A Minimal Invariant Set, Frames, and Operators

**B.S. in Mathematics,** *Florida State University* 2014 - 2017

**Minor:** Philosophy | **GPA:** 3.90

## TECHNICAL SKILLS

**Causal Inference & Experimentation:** A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, Robustness Checks

**Modeling & Statistics:** Regression, Classification, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, SHAP, Drift Monitoring

**Data & Tools:** SQL (Snowflake, SQL Server), Python (pandas, GeoPandas, statsmodels, scikit-learn, econml, MLflow), R (survey analysis), C#, Databricks, Spark, dbt, Elasticsearch, Azure, Power BI, Excel, Git

**Collaboration:** Cross-Functional Partnership (Product, Finance, Engineering, Strategy), Stakeholder Communication, Method Documentation, Mentoring

**Metrics & Planning:** KPI Frameworks, KPI / Driver Trees, Certified Metric Definitions, Leading Indicators, Acquisition & Retention Marketing Analytics, LTV / ARPU, Contribution Margin, Scenario Analysis, Target Setting, Forecasting